AIDM7020 AI and Big Data Applications in Financial Market Analytics

Final Report

Social Media Sentiment and Stock Reactions: Evidence from iPhone Launch Events (2016–2025)

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Abstract

This study examines Apple Inc. (AAPL) by using all iPhone launch events from 2016 to 2025 to build weekly event windows. Bloomberg's Twitter sentiment indicators are combined with the returns of AAPL and the S&P 500 to analyze the market impact of social media sentiment. The results show that AAPL records positive abnormal returns in most event weeks. The linear relationship between sentiment intensity and weekly returns is weak, but the direction of sentiment matches stock price movements. Lagged analysis further shows that this week's sentiment cannot predict next week's returns. In the context of "mega-cap tech stocks + product launch events + weekly data," social media sentiment functions more as a real-time indicator of market atmosphere rather than a forward-looking forecasting tool.

1. Introduction

In recent years, social media platforms such as Twitter have become essential information sources for investors, and online sentiment may influence short-term market performance (Bollen et al., 2011; Tetlock, 2007). Major events of high-profile technology companies often trigger intensive discussions on social media (Bollen et al., 2011). New product announcements in the electronics industry have been shown to induce noticeable stock market reactions (Lei et al., 2013).

Based on this, this study focuses on Apple Inc. (AAPL) and selects all iPhone launch events from 2016 to 2025 as comparable events. Weekly data are used to construct an event window covering the week before the launch, the event week, and the three weeks after. Bloomberg's Twitter sentiment indicators are then matched with the returns of AAPL and the S&P 500 to examine how social media sentiment affects stock price direction and short-term predictability. To clearly show the mechanism, this study follows four steps:

1) Test whether launch events generate systematic abnormal returns.
2) Test whether sentiment intensity affects same-week returns.
3) Examine the relationship between sentiment direction and stock movement.
4) Test whether sentiment has predictive power with a one-week lag.

2. Literature Review

Many studies show that social media sentiment can affect asset pricing in the short term, especially during major events or for highly followed companies (Bollen et al., 2011; Sprenger et al., 2014). However, its predictive ability varies across market conditions and firm types (Gu & Kurov, 2020). Event-study methods are commonly used to evaluate market reactions to major

events (MacKinlay, 1997).

Existing research still has two main gaps. First, most studies rely on daily or even minute-level data, which may not capture the broader influence of events that last several days or weeks. iPhone launch events often span long communication cycles, so weekly data may better reflect their overall effects. Second, prior studies mainly focus on sentiment intensity but pay less attention to comparing sentiment direction with market direction, although media tone is shown to have a strong influence on stock movements (Tetlock, 2007; Garcia, 2013).

To address these gaps, this study focuses on iPhone launch events and uses unified weekly event windows to compare sentiment intensity, sentiment direction, and lag effects, providing additional evidence on the synchronous and predictive roles of social media sentiment in long event windows.

3. Data and Methodology

3.1 Sample and Data Sources

To examine the role of social media sentiment in major corporate events, this study focuses on Apple Inc. (AAPL) and uses all iPhone launch events from 2016 to 2025 as the event set. The natural week of each launch event is defined as the event week, the week before, and the three weeks after, forming a five-week event window. In total, 50 weekly observations on launch events are collected for event-study analysis and regression.

Financial data comes from Bloomberg and Yahoo Finance, including AAPL's weekly high, low, and closing prices, as well as the S&P 500 weekly closing price. These data are used to calculate weekly returns and abnormal returns. To match the multi-day influence of launch events, all financial data is converted to weekly frequency.

Social media sentiment data comes from Bloomberg GT (Twitter Activity). Tweets related to AAPL are classified using a supervised machine-learning model that generates weekly statistics, including Positive Sentiment Count, Negative Sentiment Count, and Publication Count (Cui, Lam & Verma, 2016).

3.2 Variable Construction

(1) Price and Return Variables

To provide a stable measure of weekly price levels, this study uses the Typical Price, a common technical indicator (Murphy, 1999):

$$\text{Typical Price}_t = \frac{\text{High}_t + \text{Low}_t + \text{Close}_t}{3}$$

The variable t represents the week. Based on the typical price, the weekly returns of AAPL and the S&P 500 are calculated and denoted as $R_{AAPL,t}$ and $R_{MKT,t}$. Using the market-adjusted model in event-study research, the abnormal return is defined as

$$AR_t = R_{AAPL,t} - R_{MKT,t}$$

This measures the difference between AAPL's actual return and the market benchmark during the event window (MacKinlay, 1997) and is used to capture AAPL's excess performance around the iPhone launch events.

(2) Social Media Sentiment Variables

To turn unstructured social media text into numerical indicators, this study constructs a weekly sentiment score based on Bloomberg tweet counts:

$$\text{Sentiment Score}_t = \frac{\text{Positive Count}_t - \text{Negative Count}_t}{\text{Publication Count}_t}$$

The score range is [−1, 1], with higher values indicating more positive Twitter sentiment around AAPL. This standardized measure can be directly compared with weekly returns and abnormal returns.

3.3 Empirical Methodology

This study combines event-study methods with weekly time-series regressions.

First, following MacKinlay (1997), weekly abnormal returns are calculated within the event window, and the pattern of abnormal returns before, during, and after each launch is examined to evaluate short-term excess performance.

Second, two types of regressions and correlation tests are conducted to analyze the relationship between sentiment and stock performance:

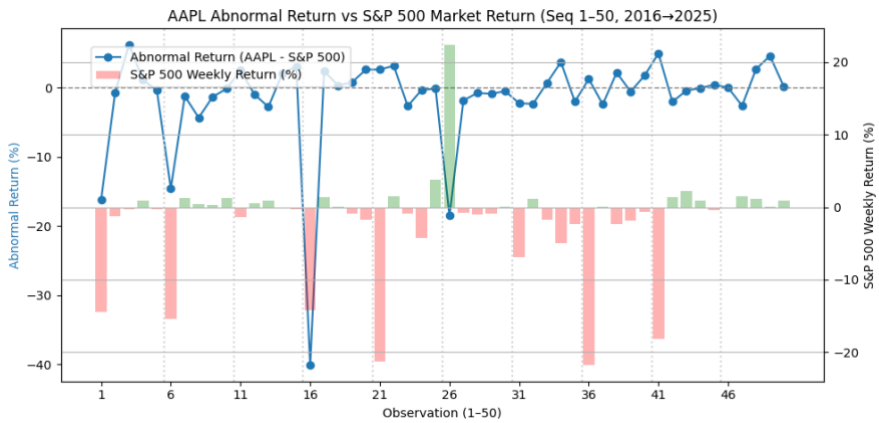
(1) Regress AAPL's weekly return or abnormal return on the same-week sentiment score to measure the contemporaneous effect;

(2) Regress next week's return on this week's sentiment score to test for one-week predictive power.

Comparing these models helps determine whether Twitter sentiment reflects immediate price reactions or contains forward-looking information.

4. Results

4.1 Abnormal Returns Around iPhone Launch Weeks



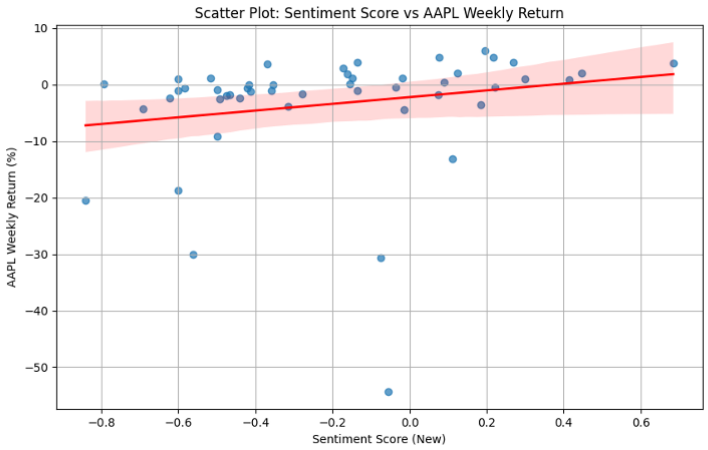
We first test whether iPhone launch events lead to market-level price reactions by examining abnormal returns. Prior studies show that new product releases often generate short-term positive abnormal returns (Lei et al., 2013). Using the five-week event window from 2016 to 2025, a total of 50 weekly observations are included.

Based on the market-adjusted model:

$$AR_t = R_{AAPL,t} - R_{MKT,t}$$

Most event weeks show positive abnormal returns, indicating that AAPL often outperforms the market during iPhone launch weeks. Some weeks have stronger reactions, and some are more moderate, but the overall trend is still positive. This general pattern supports the later analysis of social media sentiment.

4.2 Contemporaneous Social Media Sentiment and Stock Performance



After confirming market reactions, we test whether sentiment intensity affects weekly returns. Pearson correlation results from the 50 event weeks show only a weak positive correlation between sentiment score and AAPL's weekly return ($r = 0.1623–0.1940$), with p-values above 0.18, indicating no significant effect. This limited explanatory power is common for large technology firms (Gu & Kurov, 2020).

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====== 2×2 Contingency Table (Sentiment × Price Movement) ======
return_class     Down  Up_or_Flat
sentiment_class
Negative           25          10
Positive            4          10
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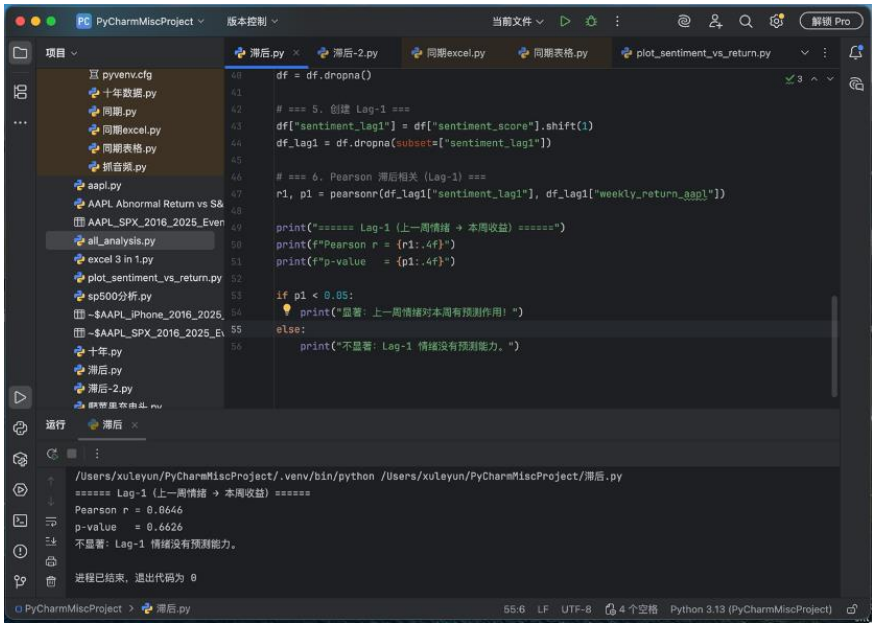
Since intensity is not significant, we focus on sentiment direction. After classifying sentiment as positive or negative and matching it with stock up/down movements, a 2×2 contingency table shows that positive sentiment aligns more often with price increases (10 times), and negative sentiment aligns more often with price decreases (25 times). A chi-square test ($\chi^2 = 5.93$, $p = 0.0149$, $\varphi = 0.3480$) indicates a significant and meaningful relationship between sentiment direction and stock direction.

Overall, sentiment intensity cannot explain return magnitude, but sentiment direction matches price direction well, reflecting market optimism or pessimism during launch weeks.

4.3 Lagged Sentiment and Short-Term Predictability



Finally, we test whether the sentiment can predict short-term movements by using a one-week lag analysis. In this step, we run a regression that uses this week's sentiment score to explain next week's return, and the results show:

- $r = -0.0374$
- $p = 0.8007$ ($p >> 0.05$)

Both values indicate no predictive power. This suggests that weekly social media sentiment reflects what the market feels at the moment, rather than predicting what will happen the following week. This result also matches earlier studies, which found that the influence of sentiment usually lasts only 1-3 days (Bollen et al., 2011; Sprenger et al., 2014).

5. Conclusion

This study examines the market impact of social media sentiment during extended event windows, using iPhone launches from 2016 to 2025 and Bloomberg Twitter indicators. The findings reveal three key insights: First, AAPL stock exhibited predominantly positive abnormal returns during launch periods, reflecting the short-term positive effect of new releases. Second, sentiment direction holds greater explanatory power than intensity, showing a significant correlation with stock price movements despite a weak linear relationship with intensity. Third, lag analysis indicates that sentiment exhibits synchrony rather than foresight, as it fails to predict the following week's returns.

Overall, within the context of "mega-cap tech stocks + long event cycles + weekly sentiment data," social media sentiment mainly reflects contemporaneous market mood rather than future price movement. This study adds to previous research based on daily data and highlights the importance of sentiment direction in weekly analysis. Future work may include multiple firms, multiple platforms, and higher-frequency data.

Reference:

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